

Liquid bridges at the root-soil interface

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Abstract

Background The role of the root-soil interface on soil-plant water relations is unclear. Despite many experimental studies proved that the soil close to the root surface, the rhizosphere, has different properties compared to the adjacent bulk soil, the mechanisms underlying such differences are poorly understood and the implications for plant-water relations remain largely speculative.

Scope The objective of this review is to identify the key elements affecting water dynamics in the rhizosphere. Special attention is dedicated to the role of mucilage exuded by roots in shaping the hydraulic properties of the rhizosphere. We identified three key properties: 1) mucilage adsorbs water decreasing its water potential; 2) mucilage decreases the surface tension of the soil solution; 3) mucilage increases the viscosity of the soil solution. These three properties determine the retention and spatial configuration of the liquid phase in porous media. The increase in viscosity and the decrease in surface tension (quantified by the *Ohnesorge* number)

allow the persistence of long liquid filaments even at very negative water potentials. At high mucilage concentrations these filaments form a network that creates an additional matric potential and maintains the continuity of the liquid phase during drying.

Conclusion The biophysical interactions between mucilage and the pore space determine the physical properties of the rhizosphere. Mucilage forms a network that provides mechanical stability to soils upon drying and that maintains the continuity of the liquid phase across the soil-root interface. Such biophysical properties are functional to create an interconnected matrix that maintains the roots in contact with the soil, which is of particular importance when the soil is drying and the transpiration rate is high.

Keywords Mucilage · Ohnesorge number · Rhizosphere · Root water uptake · Surface tension · Viscosity · Water potential

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Introduction

Root water uptake and rhizosphere biophysics

The ability of plant roots to extract water from the soil at the flow rate required to sustain the transpiration demand depends on the hydraulic conductivities of the roots, of the soil and of their interface (Passioura 1980). The relative importance of root and soil hydraulic conductivities depends on the soil water potential: the hydraulic conductivity of wet soils is higher than that of the roots, hence, it can be neglected; in dry soils the soil

hydraulic conductivity drops of several orders of magnitude becoming the limiting factor for root water uptake; at intermediate conditions, soil and root conductivities are comparable (Draye et al. 2010).

Since the pioneering work of Gardner (1960), large progress has been done in understanding plant and soil water relations. Gardner (1960) employed an analytical model to solve the radial flow of water toward the root surface to estimate the role of the soil properties in sustaining adequate water flow to the root surface. He found that large gradients in soil water potential (and soil moisture) occur around the root only when the soil is approaching the wilting point (assumed to be -1.5 MPa). The author treated the root-soil interface as a simple boundary condition, either a constant flux or a constant potential, and the soil properties were assumed to be constant – i.e. the properties of the soil close to the roots were assumed to be equal to those of the adjacent bulk soil.

However, subsequent studies have shown how complex and different from the bulk soil are the hydraulic and physical properties of the root-soil interface. Huck et al. (1970) showed that roots shrink at high transpiration rates in drying soils. Root shrinkage results in a progressive loss of contact with the soil and in a consequent reduction in hydraulic conductivity of the root-soil interface (Herkelrath et al. 1977, Faiz and Weatherley 1982). Carminati et al. (2013) observed that roots of lupine (*Lupinus albus* L.) growing in a sandy soil lost contact only after transpiration was severely reduced, concluding that air-filled gap formation at the root-soil interface was a consequence and not the cause of limited water availability to the plant roots. Probably, root hairs and the formation of stable rhizosheath around the roots helped to maintain the contact between the root surface and the soil.

Rhizosheath, which is defined as the layer of soil particles adhering to the root surface after the roots are excavated from the soil, has been shown to be strongly correlated to the presence of root hairs (George et al. 2014). An additional important element for the formation of stable rhizosheath is root exudation and in particular the excretion of mucilage (Watt et al. 1994; McCully 1999). Mucilage is a polymeric gel exuded by the root tips (Oades 1978; McCully and Boyer 1997). It is mainly composed of polysaccharides but it contains also a small fraction of lipids (Read et al. 2003).

Mucilage is believed to be responsible of unexpected soil moisture dynamics observed in the rhizosphere

(Carminati et al. 2010). Young (1995) measured higher water content in the rhizosheath of wheat (*Triticum aestivum* L.) compared to the bulk soil and interpreted this observation as the effect of mucilage hygroscopic properties. Similarly, neutron tomography experiments by Moradi et al. (2011) showed increasing water contents toward the root surface of lupine, maize (*Zea mays* L.) and chickpea (*Cicer arietinum* L.). Neutron radiography of water dynamics in the rhizosphere of lupines undergoing a drying and wetting cycle partly supported these results (Carminati et al. 2010). The authors found that during drying the rhizosphere was indeed wetter than the bulk soil. However, after rewetting, the rhizosphere remained markedly dry and it rewetted only in a few days. Again, the slow rewetting of the rhizosphere was interpreted as the effect of mucilage. This speculation was confirmed by measurements of high contact angles of the rhizosphere upon drying (Moradi et al. 2012) and hydrophobicity of dry mucilage (Ahmed et al. 2015). Subsequent experiments with mucilage analogues confirmed that mucilage increases the soil moisture during drying (Ahmed et al. 2014; Kroener et al. 2014), but it temporarily reduces the wettability of the soil after drying and subsequent rewetting (Benard et al. 2015; Kroener et al. 2015; Ahmed et al. 2016).

Although these studies have demonstrated the importance of mucilage in controlling the soil moisture dynamics in the rhizosphere, the detailed physical mechanisms of how mucilage interacts with the soil matrix and soil water hereby determining the soil hydraulic properties remain unclear.

In a recent review, Bengough (2012) asked whether the negative water potentials of mucilage measured by McCully and Boyer (1997) and Read and Gregory (1997) were primarily caused by osmotic effects (due to solutes in the mucilage) or matric effect (caused by the network matrix). A second question regards the effect of the low surface tension of mucilage compared to water. Read et al. (2003) suggested that the lower surface tension of mucilage might be functional to decrease the suction required to drain fine pores. On the other hand, the authors admitted that the easier pore drainage would decrease the unsaturated conductivity possibly limiting root water uptake in drying soils. The question concerning the role of reducing the surface tension of the soil solution remains open.

An additional open question is the impact of mucilage viscosity on water transport through the rhizosphere. As mucilage dries, its network shrinks and its viscosity

increases. The high viscosity of mucilage compared to water decreases the hydraulic conductivity of the rhizosphere, as shown by Kroener et al. (2014) and by (Rosenzweig et al. 2009) – note that in the latter case the measurements were done with bacterial biofilms. However, at some concentrations and in some soil textures, the high viscosity might be compensated by the high water holding capacity of mucilage. The effects of mucilage on the unsaturated hydraulic conductivities of soils with varying texture remain controversial (Carminati et al. 2016).

This review is focused on the physical interactions between mucilage, water and soils. Broader recent reviews on rhizosphere physical properties, including rhizosphere structure, nutrient uptake and interactions with soil microbiology, can be found in York et al. (2016), Roose et al. (2016), Oburger and Schmidt (2016) and Carminati et al. (2016). The objective of this review is to provide a theoretical framework of the physical interactions between mucilage and the soil pore matrix and the impacts on soil water dynamics in the rhizosphere. We discuss three key physical properties of root mucilage: 1) water retention, 2) surface tension and 3) viscosity. We review selected measurements of these properties and explain how their interplay and the fact that mucilage is in a porous medium contribute to the formation of an interconnect network spanning through the rhizosphere. In particular, we explain how the interplay between these three properties affects the formation and stabilization of liquid bridges between soil particles and the root surface (Fig. 1). Finally, we introduce a new conceptual model of the spatial configuration of water at the root-soil interface mediated by mucilage exudation.

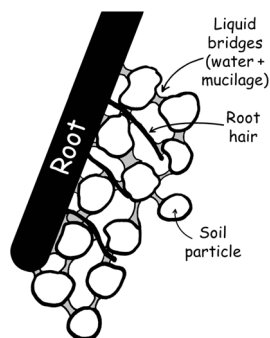


Fig. 1 Illustration of the rhizosphere and its physical components: one root with root hairs, soil particles, and liquid bridges connecting the root surface and the soil particles. The spatial configuration of the liquid phase depends on soil water potential and is affected by the presence of mucilage secreted by the root

Physical water-related properties of mucilage

Water adsorption

Mucilage can adsorb large volumes of water. Fully hydrated mucilage from axenic primary roots of maize can adsorb an amount of water a few hundred times higher than its dry weight (Guinel and McCully 1986). Ahmed et al. (2015) collected mucilage from emerging crown roots of maize and found that their full hydrated mucilage had a water content of 150 and 180 g gr⁻¹ (for plants grown in dry and wet soils, respectively). Impressed by the capacity of mucilage to adsorb such a large volume of water, McCully and Boyer (1997) hypothesized that mucilage could help plants to tolerate drought by providing an additional source of water when the soil is dry. However, the authors found that large fraction of this water could not be held against negative water potentials (Fig. 2a), concluding that mucilage per se has almost no capacity to retain water in the rhizosphere. This conclusion was reached using mucilage collected from nodal roots of greenhouse-grown maize.

Read et al. (1999) found different results using axenic maize mucilage. Their results showed higher water retention of mucilage compared to the experiments by McCully and Boyer (Fig. 2a). The differences between the two experimental results could be caused by variation in the properties of mucilage of nodal and primary roots, also reflecting a different developmental stage of the plant. Note that Ahmed et al. (2015) measured different water contents of mucilage exuded by plants exposed to soil drying, suggesting that mucilage properties might vary depending on the plant status. Another possible explanation of the differences in the measurements is that the psychrometer is not accurate at high water potentials and measurements at water potentials less negative than -100 kPa should be carefully interpreted.

Let us now calculate the effect of mucilage on the soil water retention curve. Consider a representative mucilage concentration per dry weight of soil of 0.01 (gram of dry mucilage per gram of dry soil) and a soil volumetric water content $\theta = 0.15$ [m³ m⁻³]. Assuming a bulk density of 1.5 [g cm⁻³], we find that the gravimetric water content is 0.10 [g g⁻¹] and the concentration of mucilage in the liquid phase is 0.1 (gram of dry mucilage per gram of liquid phase). According to Fig. 2a, the water potential of mucilage at a concentration of 0.1 [g

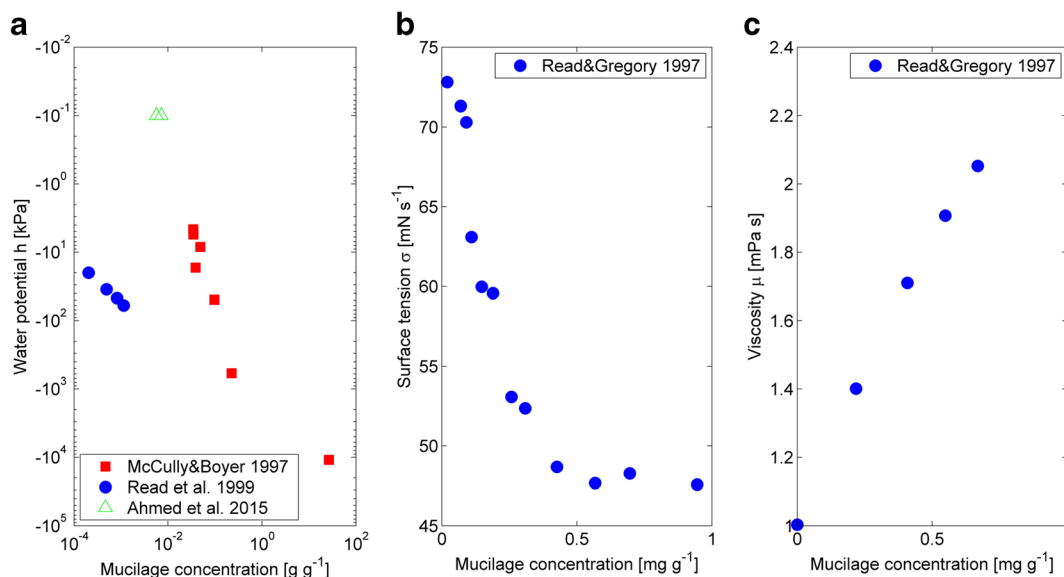


Fig. 2 Water related physical properties of the rhizosphere. a) Water potential of mucilage for varying mucilage concentrations (gram of dry mucilage per gram of water). The data are taken from McCully and Boyer (1997), Ahmed et al. (2015) (both mucilages from nodal roots of maize) and Read et al. 1999 (mucilage from

g⁻¹) is ca -100 kPa (according to the data of McCully and Boyer 1997). This means that the presence of mucilage would decrease the water potential of a soil at $\theta = 0.15$ by more than 100 kPa, thereby increasing its capacity to retain water. This approximation fits well with measured water retention curves of a sandy soil mixed with mucilage from chia seeds (*Salvia hispanica* L.). Indeed, at $\theta = 0.15$ mucilage at a concentration of 0.0125 [gr gr⁻¹] decreased the water potential of ca. 100 kPa (Kroener et al. 2014). Note that this preliminary calculation was based on the assumption that mucilage does not change the relation between water volume and matric potential. We will see later on in the text that this assumption is not always justified.

In summary, existing measurements support the hypothesis that mucilage holds water and it can increase the water retention of soils. Experiments with mucilage from different plants are needed to generalize this conclusion. Additionally, to fully understand the effect of mucilage on the soil water retention curve we need to consider also the surface tension of mucilage.

Surface tension

The surface tension determines the work needed to stretch the gas-liquid interface. In soils, the surface tension of the liquid phase is related to the curvature of

seminal roots of maize seedlings). b) Surface tension of maize mucilage for varying mucilage concentration (Read and Gregory 1997). c) Viscosity of maize mucilage for varying mucilage concentration (Read and Gregory 1997)

the gas-liquid interface according to the Young-Laplace equation:

$$h_{cap} = \sigma \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \quad (1)$$

where $h_{cap} = P_w - P_a$ [Pa] is the pressure difference between the liquid (P_w) and gas phase (P_a), σ is the surface tension of the gas-liquid interface [mN m⁻¹], and r_1 and r_2 [m] are the curvature radii of the interface (negative when the radius points towards the liquid phase). Note that in soils the gas pressure is typically equal to the atmospheric pressure, so that h_{cap} is equal to the water pressure minus the atmospheric one.

The Young-Laplace equation allows predicting the air entry-value, which is the point when air invades the pore space and the pores are drained. For cylindrical pores of radius r , the radii of curvature of the gas-liquid interface are equal to $r/\cos\alpha$, with α being the contact angle between the liquid phase and the pore wall. Then Eq. [1] becomes:

$$h_{cap} = -\frac{\sigma \cos\alpha}{r} \quad (2)$$

Eq. [2] says that in soils with small pore radii r , the capillary pressure h_{cap} required to drain the pores is very negative. In other words, much work is needed to drain small pores. Reducing the surface tension or increasing

the contact angle α lower the work needed to drain the pores – i.e. h_{cap} becomes less negative.

Root mucilage has a different surface tension than pure water and after being exuded in soil it changes the surface tension of the soil solution. Read and Gregory (1997) collected mucilage from maize and narrow leaf lupine (*Lupinus angustifolius* L.) seedlings and measured their surface tensions for varying mucilage concentrations. They found that the surface tension rapidly declined with increasing mucilage concentration, decreasing from 72 [mN m⁻¹] (surface tension of pure water at room temperature) down to 45–50 mN m⁻¹ at concentrations of 1 mg g⁻¹ (weight of dry mucilage per weight of the liquid solution) (Fig. 2b). At higher concentrations the surface tension did not decrease further. Mucilage of lupine had a slightly smaller surface tension than that of maize, but the general behavior was comparably similar. Similar properties were measured for wheat mucilage (Read et al. 2003). Read et al. (2003) attributed the low surface tension of mucilage to phospholipids. The amount of lipids in mucilage is smaller than that of sugars, which are the main mucilage components. However, phospholipids are powerful surfactants capable of reducing the surface tension of liquids even at low concentrations.

The presence of phospholipids is probably the reason of the hydrophobicity of dry mucilage (Ahmed et al. 2016; Zickenrott et al. 2016). The contact angle between water and dry mucilage increases with mucilage concentration (weight of dry mucilage per solid surface) (Ahmed et al. 2016) and it explains the strong water repellency of the rhizosphere of plants exposed to drying/wetting cycles (Moradi et al. 2012).

The implications of the low surface tension of mucilage for the soil physical properties and root water uptake were discussed by Read et al. (2003). The authors speculated that the low surface tension of mucilage facilitates the drainage of fine pores in the rhizosphere, as predicted by eq. [2]. However, the authors admitted that an easier drainage of the rhizosphere would be followed by a decrease in its unsaturated hydraulic conductivity and, eventually, by a reduced root water uptake. One might also argue that a lower surface tension of the soil solution in the rhizosphere is functional to increase oxygen diffusion and avoid anoxic conditions at the root surface. Beside these speculations, the importance and function of the low surface tension of mucilage remain unclear.

Viscosity

Viscosity expresses the resistance of a liquid to flow. When water flows in a cylindrical pore, concentric layers of water slide over each other with different velocities increasing from the pore-wall (where velocity is null due to adhesion) toward the pore center (where the velocity is maximum). The different velocity of the water layers creates an internal friction and the property of the fluid to resist the rate of shearing depends on the liquid viscosity. The flow velocity q [m s⁻¹] in a cylindrical pore of radius r and length L is described by the Pouseuille's law:

$$q = \frac{r^2 \Delta P}{8\eta L} \quad (3)$$

where ΔP [Pa] is the water pressure difference between the entrance and exit of the pore and η [Pa s] is the viscosity. For Newtonian fluids, such as water, viscosity is constant and it does not change with the shear stress. Eq. [3] says that the hydraulic conductivity of a pore is proportional to the pore radius to the power of two and it is inversely proportional to the liquid viscosity. It follows that an increase in the liquid viscosity is proportional to a decrease in soil hydraulic conductivity.

Mucilage is largely composed of polymers forming long chains. When mucilage is subjected to shear stress, the polymer chains resist to the deformation because of physical forces (the long chains are entangled, like a spaghetti dish) and chemical forces (for instance when polymers are cross-linked). Consequently, viscosity increases with mucilage concentration. Read and Gregory (1997) showed that viscosity of maize mucilage is twice that of water already at concentrations of 0.7 mg g⁻¹ (Fig. 2c). At higher concentrations, the viscosity is expected to increase further, reaching values 1000 times higher than pure water at mucilage concentrations of ca. 10 mg g⁻¹, as shown for chia mucilage in Ahmed et al. (2016). Morris et al. (1981) studied the relation between concentration and viscosity of polysaccharide solutions. They found that above a critical concentration, the polymers form a network and the viscosity started to increase more rapidly. Also, it is important to mention that mucilage, like many other gels, is not a Newtonian liquid, and it typically shows a shear thinning behavior, with the viscosity decreasing with increasing shear rate (as seen for low oscillation frequency in Read et al. 1999).

Further measurements are needed to assess differences in viscosity of mucilage from varying plant species.

The higher viscosity of mucilage compared to water causes a reduction in the hydraulic conductivity of soils embedded with mucilage. For instance, Kroener et al. (2014) observed that a sandy soil mixed with chia mucilage at a concentration of 3–5 [mg g⁻¹] (milligram of dry mucilage per gram of dry soil) had a saturated conductivity 100 times lower than the same soil without mucilage.

However, the relation between mucilage viscosity and its impact on the soil hydraulic conductivity is not as simple as in eq. [3], which states that the hydraulic conductivity is inversely proportional to the viscosity. First of all, mucilage might not occupy homogeneously the whole pore space, leaving regions where water can preferentially flow. Secondly, mucilage viscosity depends on how easily the polymer network deforms when subject to a shear stress. Changes in the polymer network definitely take place during drying and wetting cycles, when the network shrinks and swells. However, in situations when the water flow is not associated with a change in soil water content – i.e. in steady state flow – the polymer network does not necessarily deform, and the resistance to water flow depends only on the friction of the water flowing across the fixed network and not on how easily the network deforms. The examples of chia mucilage reported in the latter paragraphs supports well this point. In fact, at a concentration of 10 mg g⁻¹ mucilage is 1000 more viscous than water (Ahmed et al. 2016). This concentration in the liquid phase corresponds to a concentration in saturated soils of 3–5 [mg g⁻¹] (gram of dry mucilage per gram of dry soil) at which the saturated hydraulic conductivity is reduced by only 100 times (Kroener et al. 2014). So, the increase in viscosity (1000 times) is bigger than the decrease in conductivity (100 times).

Despite the questions on the non-linearity of the relation between conductivity and the inverse of the viscosity, the fact that mucilage reduces the hydraulic conductivity of saturated soils is quite evident. Less obvious is the effect of mucilage on the hydraulic conductivity at unsaturated conditions. In fact, in unsaturated soils two processes with opposite effect on the hydraulic conductivity take place: the first is the increase in viscosity with a consequent decrease of the hydraulic

conductivity (which was discussed above); the second is the increase of the water retention, with a possible increase of the hydraulic conductivity at a given negative soil water potential. Depending on the relative importance of the two processes, the increase in viscosity and the increase in water retention, it is not clear if mucilage increases or decreases the unsaturated hydraulic conductivity (when plotted as a function of the soil matric potential). At the moment no data are available on the unsaturated conductivity of soils mixed with varying concentrations of mucilage. Recent measurements with soils mixed with microbial biofilms (which have analogies with root mucilage), show that biofilms always reduced the hydraulic conductivity, even in dry conditions, but the relative hydraulic conductivity of soils embedded with biofilms decreased less compared to that of untreated soils (Volk et al. 2016). Further experiments are needed to confirm if such results can be extended to root mucilage.

Liquid configuration in the rhizosphere

In the next paragraphs, we propose a conceptual model of how mucilage affects the retention and spatial configuration of the liquid phase in soils. The model is based on the interplay between the three physical properties discussed above (water adsorption, reduced surface tension and increased viscosity) and on novel microscopic images of mucilage in porous media.

Liquid filaments

We used a light transmission microscope to image the spatial configuration of mucilage in soils. The images were obtained using a light transmission microscope (Olympus BX40). We prepared the samples by mixing soil particles with varying amounts of mucilage. We used different particle size, ranging from quartz sand to silty particles. We also used glass beads because we were interested in the effect of surface roughness/smoothness on mucilage deposition. Mucilage from chia seeds was used as an analogue of root mucilage. Mucilage was collected as described in Ahmed et al. (2014) and Kroener et al. (2014). Wet mucilage and sand were mixed, the mixture was placed on thin glass slides and it was let dry at ambient temperature. The drying period lasted ca. 48 h. After drying the distribution of the liquid phase was imaged using light

transmission microscope. For better visualization the samples were stained by repeated immersion into an ink/water solution, they were subsequently rinsed with deionized water and exposed to ambient humidity for ca. 48 h.

Microscopic images show that during drying mucilage is deposited preferentially in the small pores, in the contact region between soil particles and in small cavities of the soil particle rough surface (Fig. 3).

During drying mucilage formed thin filaments connecting the soil particles of coarse sand (0.5–0.63 mm in diameter, 0.036% g dry mucilage per g soil, Fig. 3a, blue arrows) and glass beads (0.1–0.2 mm in diameter, 0.086% g dry mucilage per g soil, Fig. 3c, blue arrows). The filaments were persistent and upon drying they remained visible for months. In samples of coarse sand with higher mucilage concentration (0.072%) the thickness of

the filaments increased forming large bridges between particles (Fig. 3b, green arrows). At high mucilage concentrations (0.35%) larger bridges, almost filling the complete pore space between glass beads were visible (Fig. 3d, red arrows).

In the following sections, we discuss the physical processes explaining the formation of filaments, large bridges and the observed configuration of mucilage in drying soil. Following the approach by Ghezzehei and Albalasmeh (2015) we start our analysis by considering two neighboring soil particles. Then we scale up the conceptual model to a packing of soil particles.

Interplay between adsorption, surface tension and viscosity

Consider the liquid bridge between two neighboring soil particles, which can either be in contact or

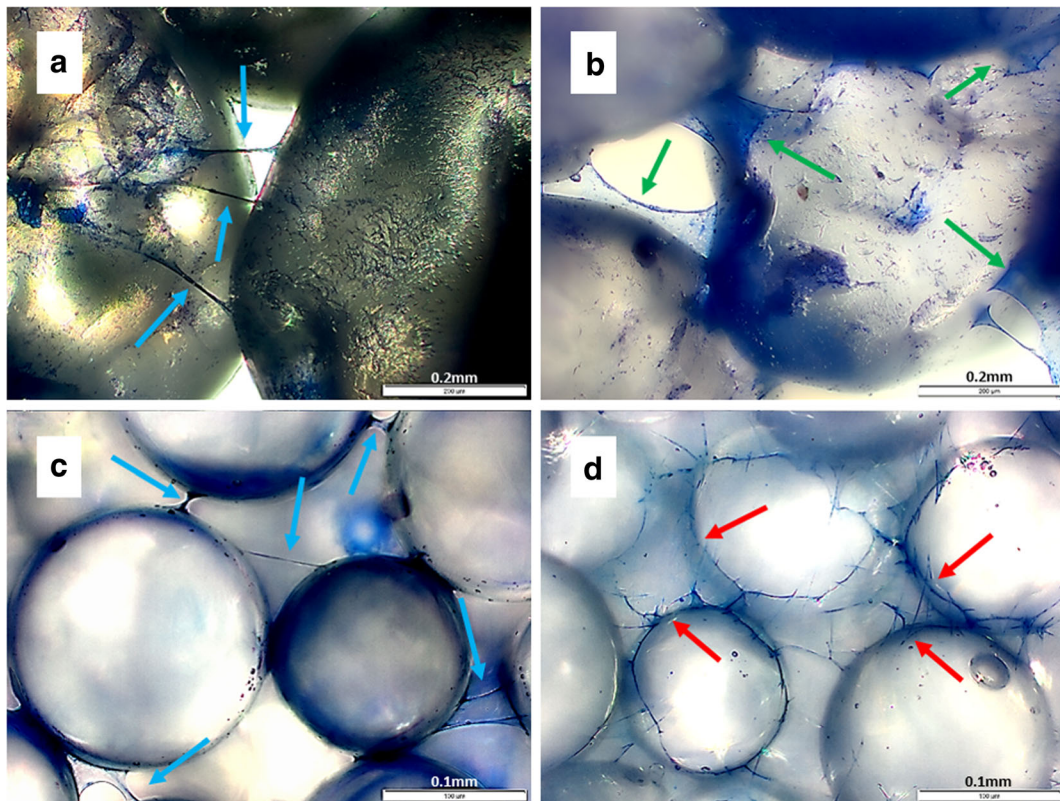


Fig. 3 Mucilage in soils visualized using a light transmission microscope (Olympus BX40). Mucilage was extracted from chia seeds, mixed with soil particles of varying diameter, let dry at ambient humidity for ca. 48 h and stained with ink. **a** Thin filaments (indicated by *blue arrows*) connecting particles of coarse sand (0.5–0.63 mm in diameter) at 0.036% mucilage concentration (gram of dry mucilage per gram of dry soil). **b** Large bridges

(*green arrows*) connecting sand particles (0.5–0.63 mm in diameter) at 0.072% mucilage concentrations. **c** Thin filaments (*blue arrows*) connecting glass beads (0.1–0.2 mm in diameter) at 0.086% mucilage concentrations. **d** At high mucilage concentrations (0.35%) large bridges fill a big fraction of the pore space between glass beads (*red arrows*)

separated by a distance L [m] (Fig. 4). The volume of liquid bridge, Vol_w [m³], depends on the radii of curvature of the gas-liquid interface (r_1 and r_2), which in turn depends on the capillary pressure h_{cap} (Eq. 1). The water potential of the liquid bridge depends on the capillary pressure as well as on the adsorption of water by the gel, which further decreases the water potential.

In this analysis we neglect adsorption of water films on the soil surface and we consider the effect of the gel to be dominant. As proposed in Ghezzehei and Albalasmeh (2015) and Kroener et al. (2014) the adsorptive potential, h_a , depends

on the concentration of mucilage in the liquid phase, which in turn depends on the water content according to:

$$h_a = -a \left(\frac{c_0 \rho_{soil}}{\theta \rho_w} \right)^b \quad (4)$$

where c_0 [g g⁻¹] is the concentration of mucilage (gram of dry mucilage per gram of dry soil), θ [m³ m⁻³] is the volumetric water content, ρ_{soil} and ρ_w [g m⁻³] are the density of the dry soil and water, and $a > 0$ and $b \geq 1$ are fitting parameters. Eq. [4] states that the drier the soil, the higher the

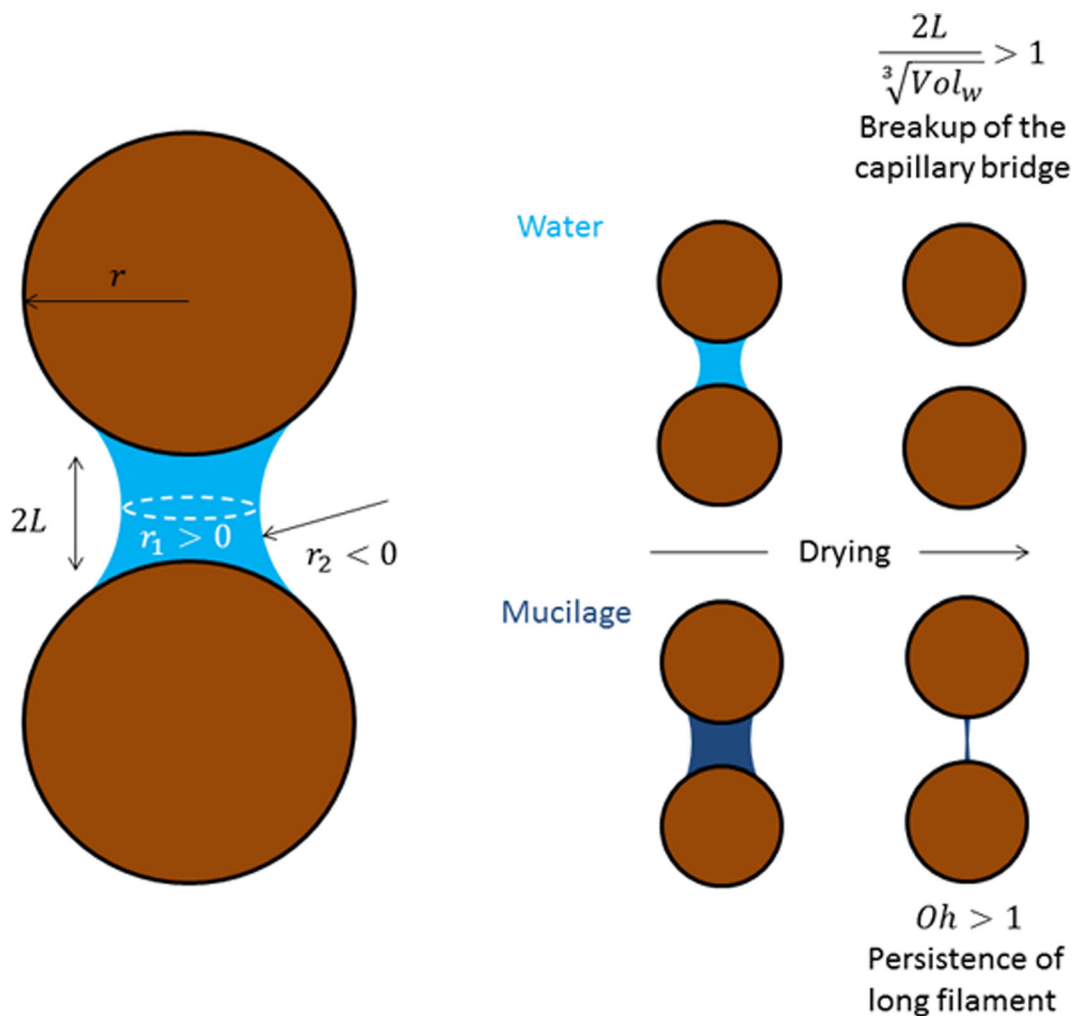


Fig. 4 Liquid bridge between one pair of soil particles having a distance of $2L$. Capillary forces retain water between the particles. At a critical volume of water, the capillary bridge breaks up (Willett et al. 2000). Mucilage decreases the surface tension of

the liquid and increases its viscosity, resulting in a high Ohnesorge number, avoiding the breakup of the liquid bridge, and allowing the formation and persistence of thin filaments connecting the two particles

concentration of mucilage in the liquid phase and the more negative the adsorptive potential.

The water potential h of the liquid bridge is the sum of the capillary and adsorptive components. From Eq. (1) and Eq. (4) it follows that:

$$h = h_{cap} + h_a = \sigma(c) \left(\frac{1}{r_1} + \frac{1}{r_2} \right) - a(c)^b \quad (5)$$

where $c = \frac{c_0 \rho_{soil}}{\theta \rho_w}$ [g m⁻³] is the concentration of mucilage in the liquid phase (gram of dry mucilage per liquid volume). Note that the surface tension also depends on mucilage concentration – i.e. $\sigma = \sigma(c)$.

Eq. [4] and [5] say that at high mucilage concentrations and in drying soils, h_a decreases reducing the decline in capillary pressure h_{cap} when h becomes negative. In other words, the adsorptive potential increases the capillary pressure at any given negative potential.

The capillary pressure and the radii r_1 and r_2 determine the volume of the liquid bridge – i.e. the water retention. Note that $r_1 > 0$ and $r_2 < 0$. For $L > 0$ (separated soil particles), when the soil dries and the volume of the bridge Vol_w decreases, r_1 becomes smaller, causing the capillary pressure to become less negative and eventually positive. At a critical point, when $\frac{2L}{\sqrt[3]{Vol_w}} > 1$, the liquid bridge breaks up (Willett et al. 2000). The reason of the breakup is that in the narrowest part of the liquid bridge (i.e. at the mid-distance between the particles) the capillary pressure is maximal and water will be drained towards the two particles, making the bridge even narrower. This situation is highly unstable and results in a quick breakup of the bridge when $\frac{2L}{\sqrt[3]{Vol_w}} > 1$.

However, the microscopic images in Fig. 3 show that the liquid bridges do not break up when mucilage is present and instead they form thin liquid filaments with small r_1 and with $\frac{2L}{\sqrt[3]{Vol_w}} \ll 1$ (Fig. 3). How to explain the persistence of these filaments?

The breakup of liquid filaments depends on the interplay between surface tension, inertial forces and viscous forces. Surface tension acts to reduce the area of the gas-liquid interface, eventually causing the breakup of the filament, but to do so it has to work against inertial and viscous forces which oppose fluid motion. The relative importance of surface tension and viscosity in the

breakup of liquid filaments is expressed by the Ohnesorge number Oh (Ohnesorge 1936):

$$Oh = \frac{\mu}{\sqrt{\rho \sigma r}} \quad (6)$$

where μ is viscosity, ρ is density, σ is surface tension and r is a characteristic length corresponding to the radius of the filament.

The Ohnesorge number (Oh) describes the damping effects of viscosity on the motion caused by surface tension. Small Oh indicates that surface tension is the dominating force, while high Oh indicates a higher importance of viscous forces. Castrejón-Pita et al. (2012) showed that long filaments never break up when $Oh > 1$.

Water has low viscosity and high surface tension, resulting in small Oh , which means that the surface tension is the dominating force and it is correct to use the Young-Laplace equation (eq. 1) to predict the shape of the air-liquid interface between soil particles and the breakup point. The presence of mucilage turns the relation around. Mucilage decreases the surface tension and it increases the viscosity. Both effects result in an increase in Oh . A preliminary calculation of Oh for a typical filament shown in Fig. 3 with surface tension and viscosity derived from Fig. 2 showed that as the water content decreases and mucilage concentration in the liquid phase c increases Oh rapidly rises approaching values much larger than 1. This explains why mucilage filaments remain intact and do not break up during drying. The lower surface tension facilitates the work to stretch the gas-liquid interface and the increase of viscosity stabilizes the liquid bridge between the soil particles. This concept is illustrated in Fig. 4.

The next step is to include the lesson learned from Ohnesorge (1936) in the computation of the retention and spatial configuration of the liquid bridges between soil particles. Our idea is to use Eq. 5 (which is based on no motion and surface tension and adsorption as dominant forces) for $\frac{2L}{\sqrt[3]{Vol_w}} < 1$. When the water content decreases further and $\frac{2L}{\sqrt[3]{Vol_w}} \geq 1$, we set the following rule: if $Oh \leq 1$, the liquid bridge breaks and its cross section goes to zero; else if $Oh > 1$ the liquid bridge does not break up and it becomes a thin filament.

We numerically computed the relation between the cross section of the liquid bridge $A_w = \pi r_1^2$ and the water potential h for varying mucilage concentration c_0 . For simplicity, we assumed that the soil particles were

spheres of equal radius, $r = 100 \mu\text{m}$, and distance $L = 10 \mu\text{m}$. We assumed that the contact angle between the liquid phase (water mixed with mucilage) and the soil particles was null. For each water configuration (determined by r_1 and r_2), we calculated the mucilage concentration in the liquid phase, the mucilage properties (surface tension and viscosity), the capillary and adsorptive potentials, and the Ohnesorge number. The functions to describe the relation between mucilage concentration with adsorptive potential, surface tension and viscosity are given in the supplementary information (Fig. S1). We made the calculations for varying mucilage concentration c_0 . The results were plotted in terms of the cross-section of the liquid bridge normalized by the particle size, $A_w^* = \frac{\pi r^2}{\pi r^2}$, as a function of the water potential h (Fig. 5). A_w^* describes the extent of the liquid connection between the two particles.

We found that at low mucilage concentrations ($c_0 < 0.01\%$ [g g⁻¹]), the liquid bridge cannot sustain very negative water potentials. As the bridge thins the water potential decreases (it becomes more negative), it reaches a minimum of ca -10 hPa, and then it starts to rapidly increase because of the decrease in r_1 . This critical point is reached just before the breakup point –

i.e. when $\frac{2L}{\sqrt[3]{V_{ol_w}}}$ approaches 1. At this stage, the liquid bridge is controlled by surface tension (capillary bridge – stage I). At higher mucilage concentrations ($c_0 > 0.01\%$), the critical point is reached at $Oh > 1$ and the capillary bridge does not break up. This means that the bridge will persist even if its water potential would be higher than that of the surrounding soil. In this case, the bridge will lose water and will narrow down to become a thin dry filament. At this stage the bridge will not break because of its high viscosity (viscous bridge – stage II). At higher mucilage concentrations ($c_0 > 0.05\%$), the capillary bridge can sustain negative water potentials. The reason is that at high mucilage concentrations the decrease in water content results in a rapid decrease of the adsorptive potential h_a , which allows to maintain a high (not so negative) capillary potential even at very negative total water potentials. In this case, the bridge is mainly dominated by the capacity of the mucilage to adsorb water (adsorptive bridge – stage III).

The parameters we used to describe the water retention capacity of mucilage, its surface tension and its viscosity (Fig. S1) are based on the properties of maize mucilage, for which a large amount of literature is

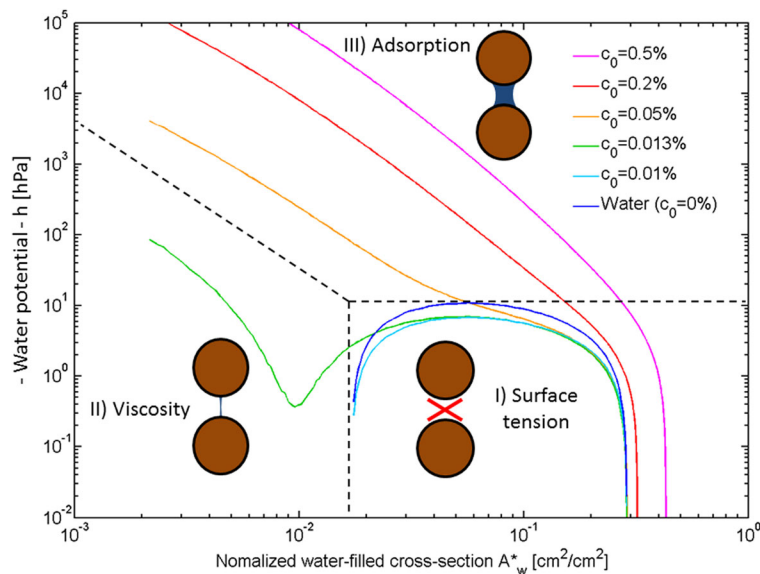


Fig. 5 Water-filled cross section (normalized for the particle cross section) of the bridge between two spheres as a function of the water potential h . Three regimes can be distinguished. At low mucilage concentrations, the capillary bridge cannot sustain very negative water potentials and breaks up (*light and dark blue lines*). At slightly higher mucilage concentrations (*green line*), the high

viscosity and low surface tension of mucilage avoid the breakup of the bridge. At low (negative) water potentials, the bridge thins and becomes a filament. At high mucilage concentrations (*orange, red and fuchsia*), large volumes water are adsorbed by mucilage also at very negative water potentials

available. A reason for it is that mucilage is easy to sample in large amounts around crown and brace roots of maize. However, mucilage from different plant species might have different properties. Our conceptual model is general enough to address differences in mucilage physical properties and it could be used to estimate the relative importance of surface tension, adsorption and viscosity for mucilage of varying plant species.

Retention and spatial configuration of the liquid phase in the rhizosphere

The study of the liquid bridge between a pair of particles is the first step to understand the retention of water in the rhizosphere. Consider now the packing of soil particles shown in Fig. 6. Two situations are discussed: low and high concentration of mucilage. The samples are

initially saturated and the water potential is gradually reduced. Air invades the pore space when the curvature of the gas-liquid interface becomes smaller than that of the inner cylindrical pore according to Eq. 2. The lower the surface tension, the less negative the matric potential needed for air to invade the pore space. Since low mucilage concentrations are sufficient to decrease the surface tension (Fig. 2b), we expect that large pores are more easily drained at low mucilage concentrations compared to pure water.

Once the large pores are drained, water is retained in smaller pores, for instance in the contact region between soil particles. As the water potential decreases, the menisci of the liquid retained between the particles recede according to the concept shown in Fig. 4, and 5. Mucilage decreases the adsorptive potential increasing the capillary potential at any given water potential,

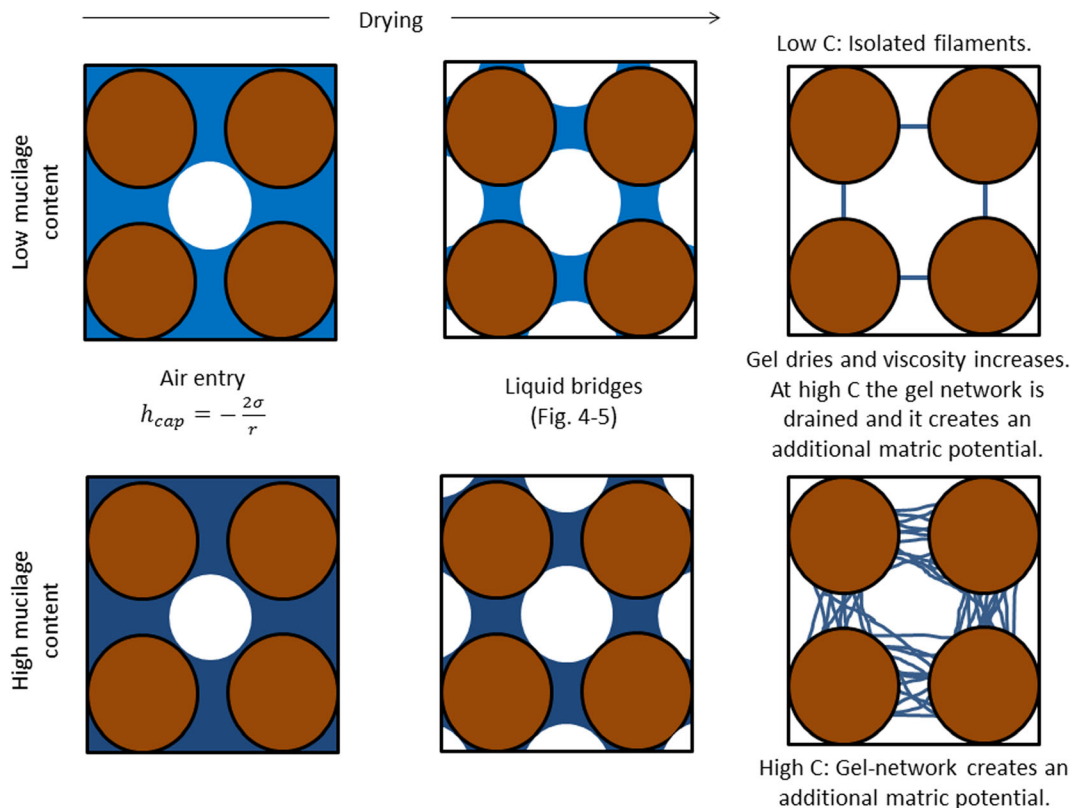


Fig. 6 Conceptual model of liquid distribution in the rhizosphere during drying at low and high mucilage concentrations. From the left to the right the samples become drier. In the upper line is illustrated a soil with low mucilage concentration, in the lower line a soil with high mucilage concentration. Air invades the inner cylinder of the pore space according to the Young-Laplace

equation (Eq. 2). The liquid bridges between neighboring particles recede according to Fig. 4, and 5. As drying proceeds further, at low mucilage concentrations the bridges shrink into thin isolated filaments (upper line). At higher mucilage concentrations, the thin filaments form a dense viscous network that creates an additional matric potential

therefore increasing the water retention. At low mucilage concentrations, filaments of mucilage form during drying because of the increasing viscosity. These filaments are isolated and the volume of water stored is small, so they are unlikely to increase significantly the water retention as well as the hydraulic conductivity of the soil.

At higher mucilage concentrations, because of the increased water adsorption, the liquid bridges remain large also at low water potentials, possibly contributing to the retention and transport of water. As the drying process advances and the water potential becomes very negative, an additional capillary potential comes into action. This additional capillary potential is caused by the network of the drying gel. As the gel dries, its viscosity increases, its mobility decreases and the gel network behaves as a fixed porous matrix with a given pore size. The network creates this capillary potential particularly when the filaments are not isolated and form a web connecting different sides of the soil particles, and for this a sufficient amount of mucilage is necessary. This additional capillary potential results from the binding of mucilage with the soil matrix and it is likely to play a smaller role when mucilage is studied in isolation. It is pretty similar to the case of a spider-web. When the spider-web is connected to opposite solid walls (like in the corner of doors and windows), the web can sustain the gravity forces that would pull it down. But when the web is disconnected from the wall it collapses. In case of mucilage alone exposed to drying, the mucilage network shrinks and the pores in network remain saturated. In case of mucilage in a porous medium, the mucilage filaments form a connected network that spans through the pore space and that is not completely free to shrink. This additional capillary potential further increases the capacity of the soil-mucilage mixture to retain water.

Implications for root water uptake

Rhizosphere properties – I mechanical stability

Drying of mucilage in soils results in the formation of long filaments that cement the soil particles together creating a stable sheath of soil around the roots. This is one of the two reasons for the formation of stable rhizosheath around roots (Watt et al. 1994). The other (well known) reason is the presence of root hairs. George et al. (2014) showed that rhizosheaths are

positively correlated to root hair density and length. Root hairs are functional to enmesh soil particles and possibly are also capable of exudation extending the zone of influence of root exudates. We believe that mucilage is similarly important for the formation of stable rhizosheaths by binding soil particles together.

The positive effect of mucilage in increasing the soil mechanical stability is caused by the formation of stable bridges between particles. Albalasmeh and Ghezzehei (2013) proposed a model of aggregate stability caused by deposition of root exudates. In their model, at a critical mucilage concentration the mucilage bridges between particle pairs are so viscous that they can be considered as solid. We took inspiration from their model, extended it to soil particle pairs that are not in contact and allowed the surface tension to vary with mucilage concentration. We then explicitly demonstrated the role of mucilage viscosity in controlling the deposition of mucilage and shaping the spatial configuration of the liquid phase in the rhizosphere.

Fig. 4 shows that low mucilage concentrations, $c_0 > 0.013\%$ [g g^{-1}], are sufficient to produce viscous bridges between soil particles (stage II). Our preliminary qualitative experiments revealed that at these concentrations soil particles are strongly cemented together. The conceptual model and our preliminary tests support the hypothesis that the long filaments increase the mechanical stability of the rhizosphere.

The high mechanical stability of the rhizosphere upon drying is functional to maintain the root and soil particles in contact. As the soil dries, roots shrink and might lose contact with the soil (Huck et al. 1970; Carminati et al. 2016). The mucilage filaments maintain the contact between the root and the rhizosphere, which is needed for water and nutrient uptake.

Rhizosphere properties – II hydraulic properties

Figure 5 illustrates how mucilage affects the liquid retention and its spatial distribution at the root-soil interface. The interplay between surface tension and water adsorption determines the water retention curve. The large reduction in surface tension already at low mucilage concentrations explains why the cross-section of the liquid bridge at $c_0 < 0.013\%$ is smaller than the one of pure water. Similarly, we expect that mucilage facilitates the drainage of large-pores at low concentrations.

At slightly higher mucilage concentrations, the combination of high viscosity and low surface tension

(elegantly summarized in the Ohnesorge number) avoids the breakup of the liquid bridges between particles. However, these bridges have a high (positive) water potential, and if connected to the liquid phase of the bulk soil would be rapidly drained and would collapse in thin filaments. Therefore, it is unlikely that the filaments per se would play a significant role in the water retention as well as in the hydraulic conductivity.

At high mucilage concentrations adsorptive forces become dominant and increase the volume of water retained in the pore space at any given water potential. The formation of a network of filaments that are not free to move (Fig. 3) (because of their high viscosity and interactions with the soil particles) may generate additional capillary forces that retain water in the pore network. In such conditions, the water retention capacity of the soil is largely enhanced, which is consistent with several experiments with soil-exudate mixtures (Kroener et al. 2014; Ghezzehei and Albalasmeh 2015) and with observations of higher water content in the rhizosphere compared to the bulk soil (Young 1995; Carminati et al. 2010; Moradi et al. 2011).

Regarding the transport of water, the effects of high mucilage concentrations on the rhizosphere hydraulic conductivity are still controversial, as discussed also in Carminati et al. (2016). Mucilage viscosity decreases the hydraulic conductivity of wet soils. In dry soils, the enhanced water retention might compensate for the higher viscosity thereby increasing the unsaturated conductivity. Possibly, the maintenance of large liquid bridges between soil particles reduces the decrease of soil hydraulic conductivity when the water potential decreases. The positive effect of mucilage might not come from an increase in the absolute value of the hydraulic conductivity, but rather from its more gradual decrease. However, this hypothesis has not been proven so far, and most of the existing measurements rather suggest a lower conductivity (Kroener et al. 2014; Volk et al. 2016). Further measurements are needed to provide conclusive evidence on the role of mucilage in shaping the hydraulic conductivity and the gradients in water potential across the rhizosphere.

Root water uptake

As roots take up water and the soil dries, the hydraulic conductivity of the soil close to the root decreases by several orders of magnitude and large gradients in water potential are needed to drive water across the

rhizosphere (Gardner 1960; Passioura 1980). The amplitude of these gradients is a function of the flow rate of water at the root-soil interface. The higher the flow rate, the larger the gradients. This relationship is highly non-linear: at a given soil moisture, there is a maximum flow rate that can be sustained by the soil. Beyond this point, either transpiration decreases or plants start wilting and their roots shrink. Furthermore, when roots shrink, they might lose contact with the soil, further decreasing the hydraulic continuity of the root-soil interface and the ability of the soil and roots to sustain the transpiration demand – a hydraulic failure!

Mucilage buffers this risk of hydraulic failure in two ways. First, the filaments maintain the roots in physical contact with the soil particles, avoiding the formation of air-filled gaps at the root-soil interface when roots shrink. Secondly, the enhanced water retention of mucilage might avoid the drop in the rhizosphere hydraulic conductivity. The latter effect might not result from an increase in the absolute conductivity, but rather from a more gradual decrease in conductivity as a function of water potential. Additionally, dynamic effects, such as non-equilibrium relation between water content and water potential, resulting from a slow drying of mucilage (Carminati 2012; Kroener et al. 2014; Schwartz et al. 2015), would maintain a relatively high hydraulic conductivity during drying events – a transitory effect. In this case mucilage would have the function to slow down and soften the drop in water potentials at the root surface. This would be very important to avoid plant water stresses during strong stress events, such as in drying soils at the peak of transpiration.

Conclusion

In this review, we discussed the effect of known physical properties of root mucilage on the retention and spatial configuration of the liquid phase in the rhizosphere. We found that the interplay between water adsorption, surface tension and viscosity determines the extent and persistence of liquid bridges between soil particles. Viscosity, which is typically ignored in predicting the stationary distribution of the liquid phase in soils, has to be considered to properly understand the formation of liquid bridges in the rhizosphere. The Ohnesorge number, which has been defined many years ago to predict the breakup of liquid jets, is useful to explain the persistence of the thin filaments forming

when mucilage dries in a porous medium. Such filaments cannot be explained using the Young-Laplace equation, which is normally used to predict water distribution in soils. At high mucilage concentrations the filaments form a network that spans through the porous medium and that creates an additional capillary force that retains water in the rhizosphere.

These considerations shed a new light on how mucilage functions in affecting soil-plant water relations. In particular, the interplay of the low surface tension and high viscosity of mucilage in shaping the spatial configuration of the liquid phase in the rhizosphere is now more evident and well described by the Ohnesorge number. Mucilage from different plant species is likely to differ in its chemical and physical properties and the conceptual framework we proposed in this review would allow to estimate the relative importance of capillarity, viscosity and water adsorption in the rhizosphere of different plant species.

Additional work is necessary to test the importance of mucilage in improving root water uptake from drying soils and in affecting soil-plant water relations. Complementary research should address the effects of mucilage in the diffusion of solutes and nutrient uptake. Experiments with genotypes with varying rates of mucilage production would be of great importance for this research field.

Finally, it would be interesting to extend these concepts to other biofilms in soils, such as EPS produced by microorganisms and exudates from mycorrhizal fungi. Interestingly, these biofilms have similar physical properties as root mucilage: they adsorb water, they are viscous and, as far as we know, many of them have a lower surface tension than pure water. It might be that these three properties are a key to understand life in soils.

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